Columns

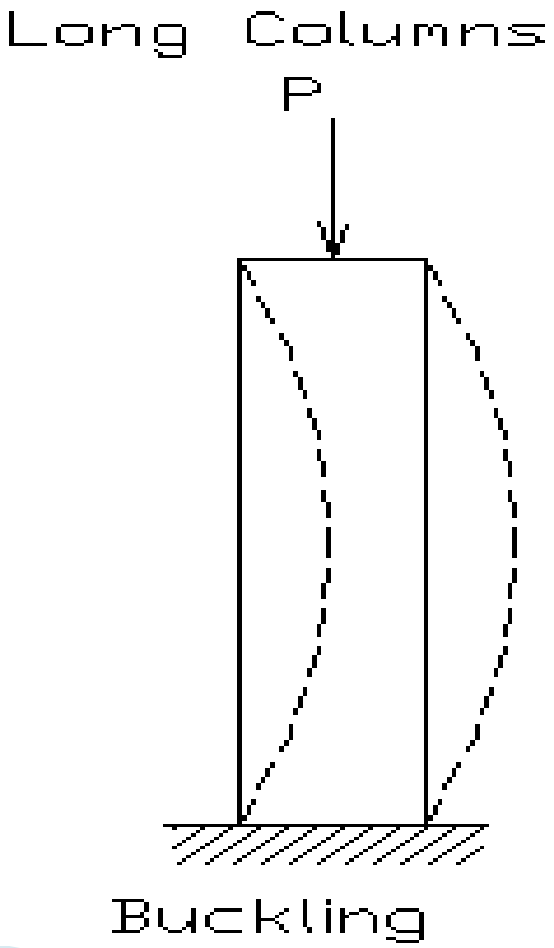
Definition:

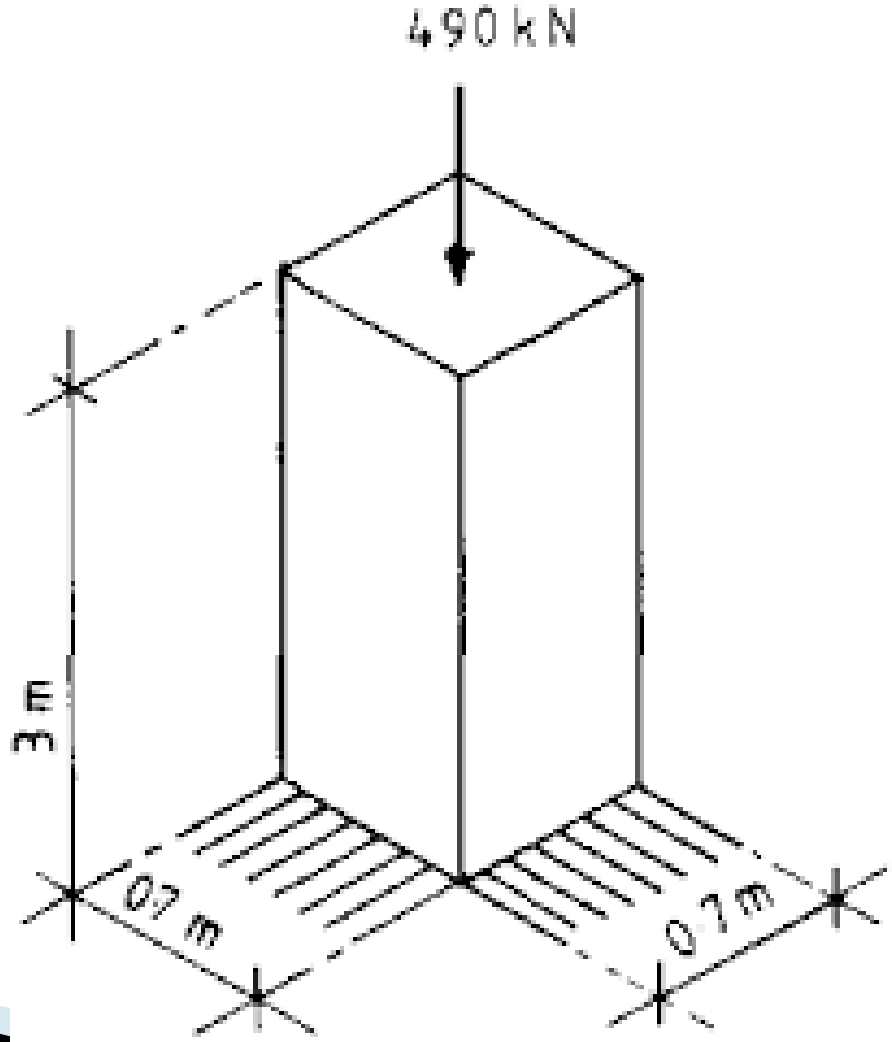
- A column is a vertical member that is loaded parallel to its axis.
- It is essentially a compressing member. Examples are vertical floor supports for buildings, support of bridges, etc

Column



Column undergoing buckling




490 kN
3 E
0.7 m
0.7 m

CLASSIFICATION

- For the purposes of design, columns can be classified as short, medium and long columns.
- The slenderness ratio is used for the classification of columns.
- The slenderness ratio (λ) is the ratio of the length (l) the column to its least dimension of the cross-section (d).

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$$\lambda = \frac{l}{d}$$

How do we know which is a short/long column?

We need to know one thing about the column.

Slenderness Ratio
(It is related to the length of a column)

Slenderness ratio $\lambda = {L_e}/{r}$

What is L_e? It is the 'length' of the column *(for a pinned-pinned end condition)*

What is *r*? It is the radius of gyration of the column.

Radius of Gyration is a term defined as, $r = \sqrt{\frac{I}{A}}$

- Short column: Slenderness ratio $\lambda < 30$
- Long column: Slenderness ratio $\lambda > 120$
- Intermediate: Slenderness ratio $30 < \lambda < 120$

It can also be defined as the ratio of the length of the column to its radius of gyration (r)

$$\lambda = \frac{l}{r}$$

$r=(I/A)^{1/2}$

I –moment of inertia, A -Area

Moment of inertia for rectangular/circular sections

$$I = \frac{a\,b^3}{12}$$

$$I = \frac{r^3}{12}$$

Effective length

- The strength of a column and the manner in which it fails depend to a greater extent on its effective length
- As the effective length of a column increases, its buckling stress will decrease
- The column is said to fail elastically if the if the buckling stress is less than the proportional limit when the effective length exceeds a certain value.

- Long columns usually fail elastically
- Where axial compressive buckling stress remain below the proportional limit

- For short columns the failure stress is equal to the yield stress
- For intermediate columns some of the fibre will reach the yield stress and some will not. The member will fail by both yielding and buckling.
- Most columns fall in this range

Failure of columns

- A short column may fail either by crushing or by shearing. For crushing, the stress is given by:

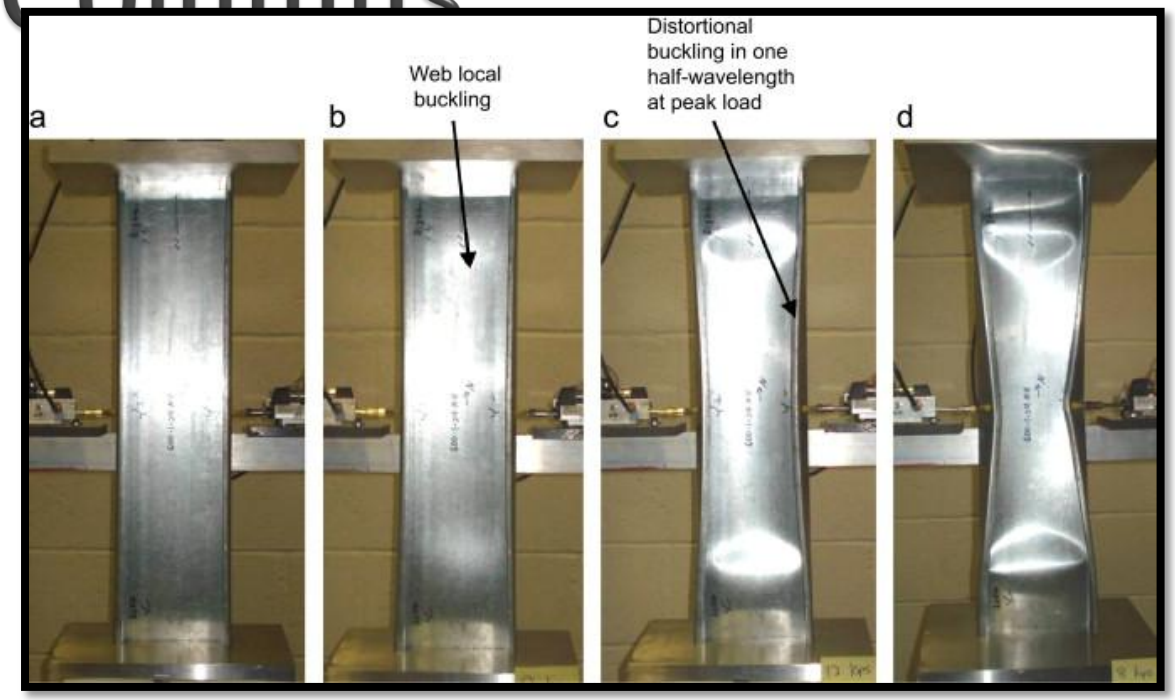
$$P = \sigma_y A$$

P = Load

σ_y = Allowable compression stress

A = Cross-sectional area

Short Columns

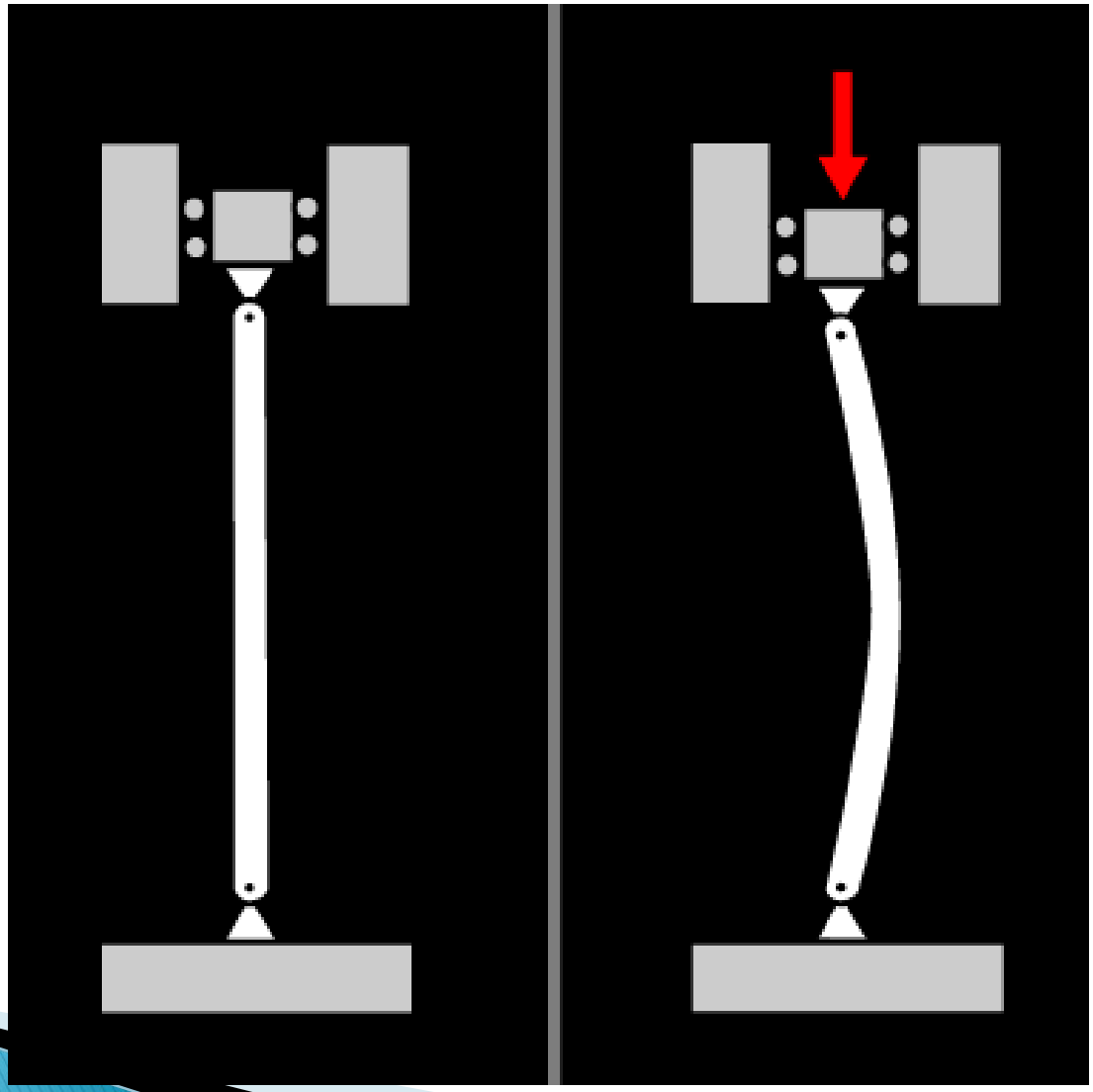


- Short columns generally don't fail by elastic buckling.
- The failure stress is close to yield stress of the material.
- The true short-columns do not have much practical application.

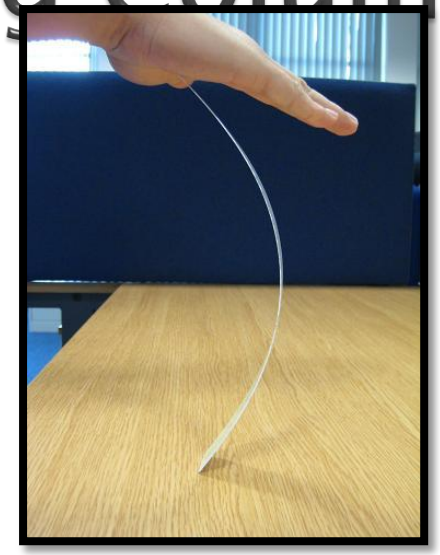
- When a **long column** is loaded axially in compression, there is a certain load at which a noticeable deflection occurs.
- The value of that load that is just sufficient to keep the column in a slightly deflected configuration is referred to as the critical load of the column.

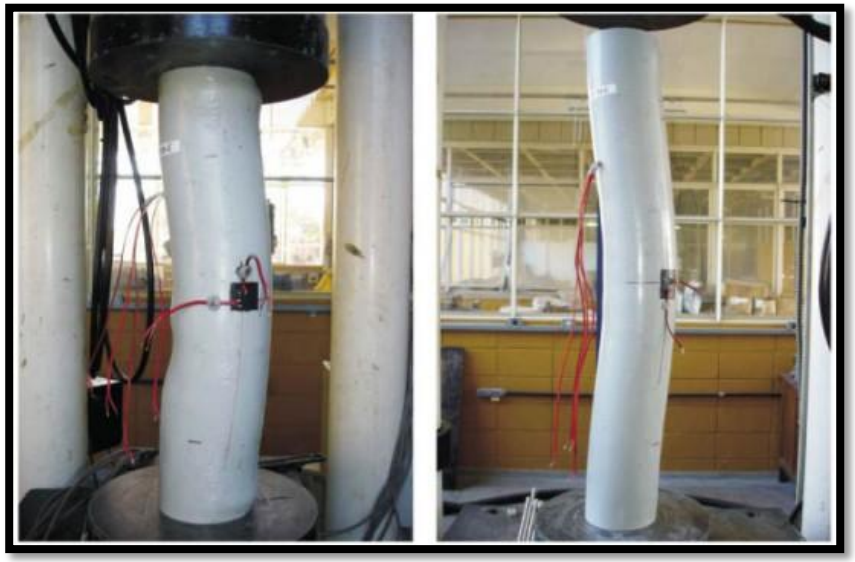
- This slightly deflected configuration phenomenon is known as **buckling**. It occurs when the stresses in the column are still below those required to cause a compression failure.

Column under buckling



Long Columns





- Long columns usually fail by elastic buckling.
- The failure load is below ultimate strength of the material.
- The Euler formula is used to calculate failure strength in long columns.

Euler's equation for Column analysis

ASSUMPTIONS

- Loads are applied at the end of the column without eccentricity
- A member or column is perfectly straight before the load is applied
- The column is made up of homogeneous material
- The weight of the material is ignored in the analysis
- The ends of the column are constructed such that the column is free to buckle at any axis of the cross-section

The critical load for a column according to Euler is given by

$$P_{cr} = \frac{\pi^2 EJ}{L^2}$$

The critical stress

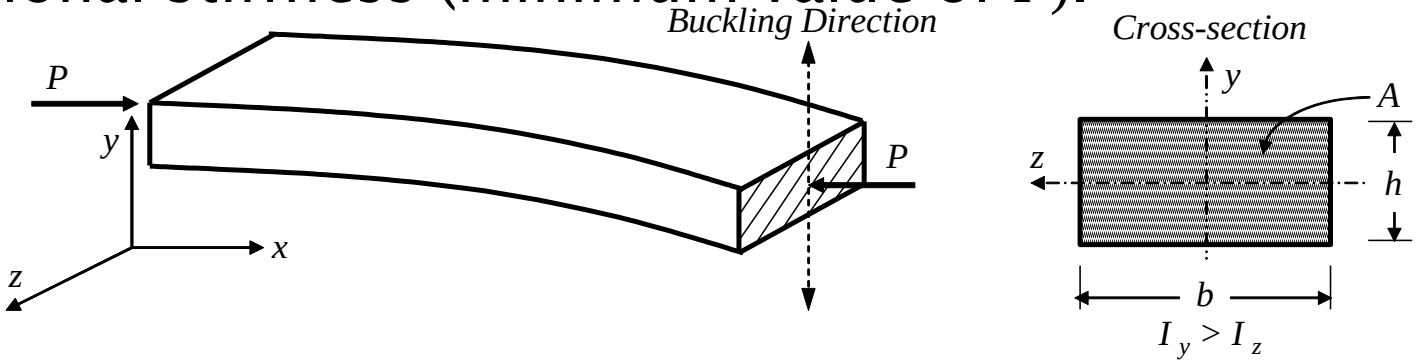
$$\sigma_{cr} = \frac{\pi^2 EI}{L^2 A} = \frac{P_{cr}}{A}$$

- For a rectangular cross-section, buckling will occur with the minimum I or r. Note that the minimum radius of gyration and for that matter the minimum moment of inertia yields the maximum slenderness ratio.
- The maximum slenderness ratio is used to determine the critical or allowable load for a column.

Cross-sectional Shape & Buckling

- *The radius of gyration provides a measure of the resistance provided by a cross-section to lateral buckling.*
- *The radius of gyration is not a physical entity in itself. It is a relationship derived to make prediction of column behaviour easy. The radius of gyration is related to the size and shape of the cross-section.*

- Columns will buckle in the direction of least cross-sectional stiffness (minimum value of I).



- *A rectangular column will buckle in the direction of the smaller dimension in cross-section. A square column cross-section will be equally prone to buckling in both x and y directions. This is because the cross section will offer equal resistance to buckling in the direction x and y.*

Application of Euler's Equation

Euler's formula may be used to determine:

- critical load of a column
- the shortest length of a column for which buckling will occur
- the critical stress
- the ratio of critical stress to load
- the relative strengths of given cross-section

Sample Questions

- A column of cross section 50mm x 60mm has a length of 2m. Determine the radius of gyration, moment of inertia and the maximum slenderness ratio.
- Write down Euler's equation for critical load and show that the critical stress is inversely proportional to the square of the slenderness ratio
- Two cross sections are given as: 50mm x 100mm and 60mm x 75mm. The two are of the same material and length. Compare their buckling strength.

A 3m column with the cross section 100mm x 125mm is constructed from timber. Determine:

- (a) the slenderness ratio,
- (b) the Euler buckling load (E= 13 kN/mm^2 for timber), and
- (c) the axial stress in the column when Euler load is applied

Example

- Determine the thickness of a round steel tubular strut, 375mm external diameter and 2.5m long, pin-jointed at the ends, to withstand an axial load of 39000kN. E=200GPa.

- Determine the slenderness ratio for a timber column 80mm X 100mm in cross-section which has effective length of 5m. This column is subjected to an axial compression load which is within the proportional limit of 8 N/mm^2. The material has E value of 8.2 kN/mm^2. Determine buckling load and the critical stress which may be used.

- A farmer requires the construction of a water tank of dimension 2m x 2m. Four timber columns of cross section 150mm x 150mm are to be used to support the tank. The timber in question has an allowable compression of 5N/mm^2 and a modulus of elasticity of 2500N/mm^2. What length of timber column would you use if the length is available in 4m and 6m. (Weight of tank =30kN and density of water =1000kg/m^3